

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Information Retrieval [IT-4035]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

1) The following two documents are in my corpus with a query:

- A) **Document 1:** "Taj Mahal is a beautiful monument."
Document 2: "Victoria Memorial is also a monument."
Document 3: "I like to visit Agra."

Query: "taj mahal agra"

(5)

Construct a Document Term Matrix (DTM) with TF-IDF values of each term and rank the documents based on the cosine similarity obtained between the query and documents.

B) We have a two-word query. For one term, the postings list consists of the following 17 entries:

[13, 15, 19, 115, 124, 139, 160, 168, 175, 181, 184, 189, 192, 196, 197, 1100, 1115]

For the other it is the one-entry postings list:

[189]

Work out how many comparisons would be done to intersect the two postings list with the following two strategies. Briefly justify your answers:

(3)

1. Using standard postings list.

2. Using postings lists with skip pointers, with a skip length of $\sqrt{p}$, where p is the length of the postings list.

C)

(2)

Given a portion of a positional index in the format:

term: {doc1: <position1, position2, ...>; doc2: <position1, position2, ...>}

lion: {1: <17, 19, 34, 73, 88, 233>;
 2: <1, 18, 94, 256, 299>;
 4: <18, 116, 1190, 1429, 1433>;
 5: <1363, 1367>;
 7: <113, 123, 1191>; ...}

roars: {1: <18, 35>;
 4: <117, 1191, 1291, 1430, 1434>;
 5: <114, 119, 1101>; ...}

Which documents do match for the given phrase query "lion roars"? Explain the process of finding the phrase queries in the given positional index.

- 2) The following three documents are in my corpus with a query:

- A) **Document 1:** "Obama rejects allegations about his own bad health."
Document 2: "The plan is to visit Obama."
Document 3: "Obama raises concerns with US health plan reforms." (5)

Query: "Obama health plan"

Estimate the probability that the above documents are relevant to the query.

- B) Draw the inverted index that would be built for the following document collection.

- Document 1:** "new home sales top forecasts."
Document 2: "home sales rise in July." (3)
Document 3: "increase in home sales in July."
Document 4: "July new home sales rise."

- C) When using weighted zone scoring, is it necessary for all zones to use the same Boolean match function? Justify. (2)

- 3) Consider that there are 8,00,000 documents, and each docID of the postings list is of size 32 bits.

- A) 1. What is the basic compressed bit size for storing docID in the postings list? (1M)
 2. Is it possible to reduce it further? (1M) (5)
 3. Illustrate the concept of variable length encoding for the docIDs < 283047, 283154, 283159, 283202> (3M)

- B) Compute the singular value decomposition of a given matrix of rank 2.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \quad (3)$$

- C) Illustrate the advantages of inexact top-K retrieval system. (2)
- 4) Compare HITS with page rank in detail. (4)
- A)
- B) Consider the strings [systile, syszygentics, syzygial, syzygy, szaibeltyite, Szczecin, szomo] are the part of dictionary made up with 400,000 terms.
1. Explain how compression will happen using dictionary-as-a-string method. (3)
2. Draw a neat sketch that demonstrates the same.
- C) The following list of Rs and Ns represents relevant (R) and nonrelevant (N) returned documents in a ranked list of 10 documents retrieved in response to a query from a collection. Assume that there are 5 relevant documents in total in the collection. (3)
- R N R N N R N N R R
- What is the interpolated precision at 0.5 recall?
- 5) Assume that the vocabulary is {lazy, fox, dog, cat, jump}. TF-IDF values for the original query $q_0 = [0, 1, 0, 1, 1]$, relevant document $d_R = [2, 1, 1, 0, 2]$ and non-relevant document $d_N = [3, 0, 1, 0, 2]$. Find the modified query q_m using Rocchio algorithm. (5)
- A)
- B) An IR system returns 8 relevant documents and 10 nonrelevant documents. There are a total of 20 relevant documents in the collection. What is the precision of the system on this search. Also, compute recall and F-1 measure for the same. (3)
- C) Develop the algorithm for Block Sort-based Indexing (BSBI). (2)

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